Engineering the Customer into the Solution

BLUE LINE INNOVATIONS

The PowerCost Monitor consists of two components: a sensor unit that attaches to the outside of the electric meter and a portable in-home display unit that shows the customer how much electricity the household is consuming.

Most conservation and demand-side management managers can agree that conserving energy is much more cost-effective than securing additional energy resources. Consequently, various innovative conservation and pricing strategies are being tested and adopted by utilities as they seek to lower costs and conserve energy. Blue Line Innovations offers utility customers an innovative tool that encourages energy conservation and educates residential customers on the cost of running their homes.

This tool is called the PowerCost Monitor; it is an in-home display device that shows customers how much energy their home is using from moment to moment, in the hope that this information will change their consumption behavior and ultimately reduce their electricity usage—an incentive that both the customer and the utility can appreciate.

So does real-time feedback on energy usage actually lead to lower electricity usage?

Yes, and not only do customers save after initial use, but they continue to conserve energy over a long period of time.

In 2007, Blue Line Innovations completed a 30,000-unit deployment to Hydro One Networks residential customers (winning Blue Line Innovations an award for Outstanding Energy Efficient Technology Deployment of the Year from the Association of Energy Services Professionals). This deployment was the result of a successful pilot with 500 Hydro One customers over a two-and-a-half-year period, from 2004 to 2006, one of the longest pilots completed on real-time feedback.

The pilot showed an average energy savings of 6.5%, enough savings to justify rolling out a full-scale program. "That to us was a very important result," says David Curtis, director, Business Transformation. "Not only did customers act on what the monitor was telling them, but they acted on it over a period of time."

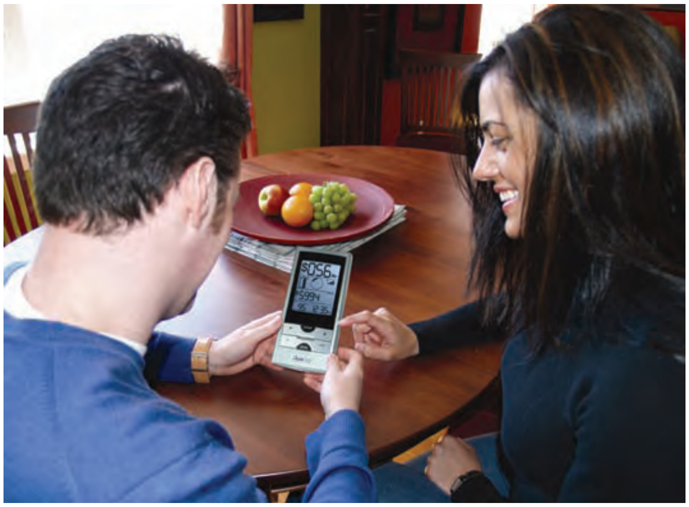
Working with a number of utilities, Blue Line Innovations has developed a business process model that helps utilities put the PowerCost Monitor into practice for its residential customers, minimizing time and money invested and lessening the impact on their core utility business while improving the success of the conservation initiative.

NSTAR Electric and National Grid both implemented pilot programs this summer to test the technology over a period of time with their respective residential customer bases. "With electricity, unfortunately, the bill comes a month later," says Tom May, CEO, NSTAR Electric. "So each and every day, as you're using energy, you're not really getting the instant message back in real time what you are doing. We are the first utility in the country offering [the PowerCost Monitor] to our customers. With this simple device … customers can see what they are spending in real time."

The PowerCost Monitor consists of two components: a sensor unit that attaches to the outside of the electric meter and a portable in-home display unit that shows the customer how much electricity the household is consuming. The PowerCost Monitor is designed to provide consumers with moment-to-moment information on their electricity consumption.

The in-home display is simple to use and easy to program. It is also compatible with 99% of residential, nondemand meters, and it can incorporate various rate structures, such as time-of-use rates, tiered rates, and volume pricing. This is great news for utilities that want to test and educate their residential customers on new pricing strategies.

Provision of feedback translates into ratepayer savings in three primary areas, all related to subtle, but persistent changes in behavior. These are adjusting the indoor temperature while actively occupying the home, or while sleeping, or while out of the home; using appliances and lighting; and making improvements to the envelope of the home.



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The PowerCost Monitor acts like a speedometer, showing the current rate of energy consumption. It also acts like an odometer, showing the cumulative consumption, that is, the total energy used to date. This direct, immediate information feedback benefits the ratepayer in two ways:

1. It makes learning the difference between energy waste and energy efficiency inexpensive, simple, and quick. Each ratepayer can make individual, educated decisions about his or her lifestyle choices and use of energy.

2. It provides positive reinforcement by confirming actual savings online from adding insulation, weatherstripping, and window shades; or from shifting time-of-use of appliances.

Pilot projects and academic studies show that when consumers have direct information feedback, they save money on their energy bills. Itemizing electricity costs in the home helps customers understand how they can save energy, and having that constant feedback acts as a behavior modification tool, allowing customers to change their energy habits and see the immediate cost savings on their in-home display.

After all, you can't manage what you can't measure.

According to Danny Tuff, CEO at Blue Line Innovations, "Our goal is to provide utility customers with real-time feedback devices that promote energy conservation. The essential premise is that efficient use of energy is the most cost-effective, the most reliable, and the most beneficial environmentally sensitive conservation method."

—James McMillan

James McMillan is vice president of marketing at Blue Line Innovations.

For more information:

For more on Blue Line Innovations and the PowerCost Monitor, go to www.bluelineinnovations.com.